Geometric Non-Stationarity of the Type Ia Supernova Standardization Manifold and Its Impact on Dark Energy Inference

Humza Hafeez*
Independent Researcher; ORCID: 0009-0000-0853-8807
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We find that the covariance structure of the $(x_1, c, \Delta\mu)$ standardization space of Type Ia supernovae is strongly redshift-dependent (Box's M test, $p = 2.3 \times 10^{-7}$), that the residual coupling to color reverses sign across the Pantheon+ redshift range ($\rho = -0.785$, $p = 3\times10^{-30}$), and that adding two interaction terms ($z \times x_1$, $z \times c$) to the standard Tripp equation yields $\Delta\chi^2 = 269$ for 1,590 SNe. Constant-basis SN Ia standardization is therefore geometrically inconsistent with the data. The 8.2σ residual–redshift anomaly is consistent with projection error from applying fixed-basis standardization to a nonstationary manifold. When the w_0w_a dark energy parameterization is refit under corrected standardization, the best-fit w_a shifts from -1.88 to -0.02, and ΛCDM plus nonstationary Tripp outperforms w_0w_a plus standard Tripp by $\Delta\chi^2 = 37$. We further show that 71% of the host-galaxy mass step is mediated through the color channel. Independent quasar and FRB analyses show qualitatively similar channel-selective degradation, suggesting that the SN anomaly may not be unique to Pantheon+. These results imply that SN-driven evidence for time-varying dark energy should be revisited under basis-aware standardization.

INTRODUCTION

The Tripp standardization [1]

$$\mu = m_B + \alpha\, x_1 - \beta\, c - \gamma\, \delta_M - M_B \tag{1}$$

has been the foundation of SN Ia cosmology since the discovery of cosmic acceleration [2, 3]. It assumes that the light-curve parameters x_1 (stretch) and c (color) span a *fixed* coordinate system whose mapping to the distance modulus is redshift-independent. Current SN-based constraints on the dark energy equation of state, and combined analyses incorporating Pantheon+ [4, 5], DES-SN5YR [6], and DESI BAO [7], rely on this assumption.

We demonstrate that constant-basis Tripp standardization is geometrically inconsistent with the Pantheon+ data. The covariance structure of the standardization space evolves with redshift (Box's M test, $p < 10^{-6}$), the channel–residual couplings reverse sign, and the standardization manifold warps progressively away from the fixed Tripp basis. This paper is empirical: we test whether the SN standardization map itself evolves with redshift and how that affects cosmological inference, without committing to a unique physical mechanism.

Throughout this work we adopt flat ΛCDM with $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.3$.

DATA AND THE STANDARD TRIPP ANOMALY

We use the Pantheon+ compilation [4] with quality cuts ($0.01 < z_{\rm HD} < 2.5$, $|x_1| < 5$, $|c| < 0.5$, $\sigma_{m_B} < 1.0$, $0 < \log M_* < 15$), retaining $N = 1{,}590$ SNe. Under standard Tripp (Eq. 1), the best-fit parameters are $\alpha = 0.145$, $\beta = 2.71$, $\gamma = -0.046$, $M_B = -19.315$, with $\chi^2/\nu = 13.6$. Because these exploratory fits use diagonal uncertainties rather than the full Pantheon+ covariance defined in post-standardization μ-space, absolute goodness-of-fit values should not be compared directly to official Pantheon+ cosmology fits.

The Spearman residual–redshift correlation is

$$\rho(z, \Delta\mu) = -0.199, \quad p = 1.1 \times 10^{-15}\,. \tag{2}$$

From 1,000 ΛCDM mock catalogs (Gaussian noise on m_B, fixed x_1, c, M_*), the null distribution is $\rho_{\rm null} = 0.000 \pm 0.024$. The observed ρ lies at 8.2σ. The anomaly is statistically significant under this diagonal-noise mock construction.

GEOMETRIC PROOF OF NON-STATIONARITY

We establish the geometric inconsistency of constant Tripp through three independent statistical tests, none of which depend on small-bin PCA or eigenvector stability.

Box's M Test: Covariance Homogeneity

Box's M test [8] assesses the null hypothesis that the covariance matrices of the normalized $(x_1, c, \Delta\mu)$ space are equal across redshift groups. We apply it with median, tercile, and quartile splits:

Split	$M_{\rm corr}$	df	p
Median ($k = 2$, $n = 795$)	37.9	6	1.2×10^{-6}
Tercile ($k = 3$, $n = 530$)	54.5	12	2.3×10^{-7}
Quartile ($k = 4$, $n = 398$)	58.4	18	3.7×10^{-6}

The covariance matrix of the standardization space changes with redshift at $p < 10^{-6}$ across all splits. The Tripp equation assumes a fixed covariance structure; the data reject this assumption.

Rolling Channel–Residual Correlations

We track the Spearman correlation between each light-curve parameter and the Hubble residual using a sliding window of 200 SNe (Fig. 1). The color–residual coupling evolves from $\mathrm{corr}(c, \Delta\mu) = +0.09$ at low z to -0.44 at high z, with trend significance $\rho = -0.785$ ($p = 3.1 \times 10^{-30}$). The stretch–residual coupling evolves in the opposite direction ($\rho = +0.438$, $p = 7 \times 10^{-8}$). These trends are stable across window sizes of 150, 200, and 300.

The sign reversal of the color–residual coupling is a direct, continuous manifestation of the manifold's nonstationarity: the Tripp color correction, calibrated predominantly at low z, systematically over- or under-corrects at high z as the underlying relationship between color and luminosity warps.

Interaction Terms: The Simplest Test

We augment the standard Tripp equation with redshift interaction terms:

$$\mu = m_B + (\alpha_0 + \alpha_1 z)\, x_1 - (\beta_0 + \beta_1 z)\, c - \gamma\, \delta_M - M_B\,. \quad (3)$$

Term	$\Delta\chi^2$	dof	p
$z \times x_1$ alone	165.5	1	≈ 0
$z \times c$ alone	129.6	1	≈ 0
Both	268.7	2	≈ 0

Because the Tripp formalism is a linear projection from (x_1, c) into standardized distance, significant $z \times x_1$ and $z \times c$ terms directly reject the assumption of a redshift-stationary projection basis. The best-fit slopes are $\alpha_1 = -0.059$ and $\beta_1 = -0.509$: both stretch and color lose standardization power with increasing redshift, with color degrading nearly an order of magnitude faster. Each single interaction term individually rejects constant coefficients at a level exceeding any conventional threshold.

ILLUSTRATIVE GEOMETRY: PCA DRAINAGE

To visualize *what* the nonstationarity looks like geometrically, we perform PCA on the normalized $(x_1, c, \Delta\mu)$ space in five redshift bins (Table I). At $z < 0.06$, 76% of the leading variance component resides in the Tripp (x_1, c) plane. By $z > 0.5$, this drops to 50%: half of the leading variance component lies outside the Tripp (x_1, c) plane. The Grassmannian angle between the low-z and high-z PC1 subspaces is 73.4° (3.1σ above the null expectation of 24.9°).

This drainage illustrates the mechanism behind the formal tests: as the manifold warps with redshift, the fixed Tripp projection captures a progressively smaller fraction of the dominant variance, and the remainder leaks into structured residuals.

TABLE I. Fraction of PC1 variance in the Tripp plane and the residual dimension.

z-bin	N	% in (x_1, c)	% in $\Delta\mu$	Angle
$[0.01, 0.06)$	559	76.0	24.0	$-40.7°$
$[0.06, 0.11)$	78	95.3	4.7	$+39.7°$
$[0.11, 0.22)$	248	94.9	5.1	$-49.0°$
$[0.22, 0.50)$	487	57.1	43.0	$-41.8°$
$[0.50, 2.30)$	210	50.0	50.0	$-60.4°$

FIRST-ORDER CORRECTION

Evolving Coefficients and Quadratic Color

As a higher-order empirical repair, we allow all Tripp coefficients to evolve linearly and add a quadratic color term:

$$\mu = m_B + \alpha(z)\, x_1 - \beta(z)\, c - \delta\, c^2 - \gamma(z)\, \delta_M - M_B\,, \quad (4)$$

yielding $\alpha(z) = 0.148 - 0.032z$, $\beta(z) = 3.11 - 0.62z$, $\gamma(z) = -0.028 - 0.070z$, and $\delta = +3.64$. This improves over standard Tripp by $\Delta\chi^2 = 601$ for five parameters. The residual–redshift correlation decreases to $\rho = -0.135$ (32% reduction, confirmed at 4.6σ by mock null injection).

The quadratic color term ($\Delta\chi^2 = 339$) indicates that the color–luminosity relation is curved, consistent with color encoding at least two latent variables (intrinsic SED variation and line-of-sight reddening). This curvature itself evolves: $\delta(z) = 3.51 + 1.97z$ ($p = 0.0014$).

Mass Step Decomposition

The standard Tripp mass step is $\gamma = -0.046$ mag. A stretch-only fit yields $\gamma = -0.014$ mag: **71% of the mass step is color-channel driven**. This suggests that most of the observed mass step in this sample is mediated through the color channel [9–11], consistent with a population where color–luminosity coupling varies with host properties.

COSMOLOGICAL IMPLICATIONS

Absorption of Apparent Dark Energy Evolution

Under the $w_0 w_a$ parameterization [12, 13]:

Model	w_0	w_a	χ^2
Standard Tripp + w_0w_a	-0.559	-1.883	9,811
Modified Tripp + w_0w_a	-0.659	-0.020	—
Modified Tripp + ΛCDM	$w = -1$		9,775

The best-fit w_a shifts from -1.883 to -0.020. ΛCDM with nonstationary standardization outperforms w_0w_a with constant Tripp by $\Delta\chi^2 = 37$. This is a direct consequence of the geometric failure: the w_0w_a parameterization is mathematically degenerate with redshift-dependent standardization corrections. When the manifold warping is accounted for, the cosmological free parameters have no residual structure left to absorb.

Stretch-Only H_0

Removing the color correction shifts H_0 from 73.0 to 70.5 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Incorporating $\alpha(z)$ evolution reduces it to $H_0 \approx 68.8$. The direction and scale of this shift move the SN inference closer to the Planck CMB value. The color contribution to H_0 grows monotonically with redshift ($\rho = -0.800$), consistent with the progressive warping of the color channel away from the Tripp basis.

CROSS-DOMAIN EVIDENCE

Independent analyses of 750,414 SDSS DR16Q quasars [14] and 535 CHIME FRBs [15] show qualitatively similar channel-selective coupling degradation with path length: diagnostic correlations collapse at sigmoid thresholds while kinematic observables remain immune. These three source classes share no pipeline, instrumentation, or selection function, suggesting that the SN anomaly may not be unique to Pantheon+. A detailed unified analysis is reserved for a companion paper.

DISCUSSION

The central result of this work is geometric, not cosmological. We have shown—through Box's M, rolling correlations, and interaction terms—that the mapping from (x_1, c) to SN Ia residuals is redshift-dependent: the standardization manifold is nonstationary. This is a measurement, not a model. No physical mechanism is required to establish it; the failure follows from the mathematical properties of a fixed projection applied to a signal whose covariance structure evolves.

The physical origin of the nonstationarity remains open. One candidate is a propagation effect in which expanding spacetime preferentially degrades correlations between observables in proportion to the number of independent coupling channels through which information can be lost. Color (sensitive to dust, metallicity, SED shape, and reddening law) degrades faster than stretch (primarily sensitive to ^{56}Ni mass), producing the observed hierarchy. The cross-domain recurrence in quasars and FRBs supports a propagation-level origin, but a definitive derivation is beyond the scope of this empirical report.

The practical implication is immediate: dark energy constraints from SN Ia carry a geometric systematic that constant-coefficient analyses do not model. SN-driven dark energy inferences in combined analyses, including the DESI collaboration's report of $w_a = -0.75 \pm 0.29$ [7], should be revisited under basis-aware standardization before being interpreted as evidence for dynamical dark energy.

LIMITATIONS

1. **Residual floor.** The corrected residual correlation ($\rho = -0.135$, 7.0σ above the diagonal null) implies the linear repair is incomplete. The remaining structure likely reflects manifold curvature and saturation at $z > 1$, where the $[1.0, 1.5)$ bin shows over-correction.

2. **Covariance mismatch.** The public Pantheon+ covariance is defined in post-standardization μ-space [5, 16], whereas the present analysis probes nonstationarity in the upstream (m_B, x_1, c, δ_M) coordinate space. A full covariance-consistent refit therefore requires parameter-level propagation not available in the public release.

3. **DES comparison.** DES-SN5YR BBC-corrected residuals show $\rho = -0.018$, consistent with sophisticated bias-correction pipelines implicitly compensating for basis nonstationarity [6]. The geometric framework presented here provides a principled explanation for *why* such empirical corrections are necessary. Raw (x_1, c) values are not publicly available for direct comparison.

4. **Jacobian propagation.** A formal Jacobian propagation from the upstream (m_B, x_1, c) correction space into the post-standardization μ-space covariance is planned for a follow-up analysis using the SALT2 parameter-level covariance.

5. **Box's M sensitivity.** Box's M can be sensitive to departures from multivariate normality. We mitigate this by presenting it alongside the rolling correlations and interaction terms, which make no distributional assumptions.

6. **Cross-domain quantification.** The quasar and FRB analyses are qualitative. A unified fit with

a common coupling function is reserved for future work.

7. **Nature of nonstationarity.** Because the rolling-window and covariance-based tests are descriptive of nonstationarity, they do not by themselves uniquely determine whether the underlying geometry is best described as pure rotation, curvature, scale change, or a combination thereof.

PREDICTIONS

1. **Gravitational-wave standard sirens.** GW distances involve no electromagnetic standardization. We predict zero coefficient evolution and no manifold nonstationarity in GW Hubble diagrams.

2. **DESI stretch-only reanalysis.** Stretch-only standardization of the DESI SN sample should yield $H_0 \approx 68.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and substantially reduce the reported w_a deviation.

3. **Euclid eigenvector evolution.** The Euclid SN sample should show directly measurable covariance nonstationarity (Box's M $p < 10^{-3}$) and rolling-correlation sign reversal between $z = 0.5$ and $z = 1.5$.

4. **Future SN analyses with raw (x_1, c).** Any survey releasing raw light-curve parameters should show the same interaction-term significance and rolling-correlation sign reversal reported here.

CONCLUSIONS

1. The covariance structure of the SN Ia $(x_1, c, \Delta\mu)$ space is redshift-dependent (Box's M, $p = 2.3 \times 10^{-7}$). The color–residual coupling reverses sign ($p = 3 \times 10^{-30}$). Two interaction terms yield $\Delta\chi^2 = 269$. Constant Tripp standardization is geometrically inconsistent with the data.

2. The 8.2σ residual–redshift correlation is consistent with projection error from applying a fixed-basis standardization to a nonstationary manifold.

3. Under nonstationary standardization, the best-fit w_a shifts from -1.88 to -0.02, and ΛCDM + corrected Tripp outperforms $w_0 w_a$ + standard Tripp by $\Delta\chi^2 = 37$.

4. 71% of the host-galaxy mass step is color-channel driven.

5. Qualitatively similar coupling degradation in quasars and FRBs suggests the SN anomaly may not be unique to Pantheon+ and motivates broader investigation.

6. The residual is not yet white noise ($\rho = -0.135$), pointing to higher-order manifold curvature.

The simplest interpretation is that a substantial part of the apparent cosmological signal originates in a nonstationary standardization map rather than in the background expansion history itself.

This work uses the Pantheon+ compilation [4, 5], SDSS DR16Q catalog [14], and CHIME/FRB Catalog 1 [15]. Computational analysis used NumPy, SciPy, and custom Python scripts at `https://github.com/Chaiwalah/closure-theory`.

* humza@closure-theory.org

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